

Algebra II Algebra Review Cheatsheet

Module 1

Equation Solving

$$a(x + b) + c = dx + e$$

Distribute, combine like terms, collect variables, collect constants, divide by the coefficient.

Inequalities

$$-ax + b < c$$

Flip the inequality sign when multiplying or dividing by a negative number.

$$|A| < c \Rightarrow -c < A < c$$

$$|A| > c \Rightarrow A < -c \text{ or } A > c$$

Factoring Toolkit

$$a^2 - b^2 = (a - b)(a + b)$$

$$a^2 \pm 2ab + b^2 = (a \pm b)^2$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

- Factor out the GCF first.
- Use grouping for four terms.
- For $ax^2 + bx + c$, look for factors whose product is ac and sum is b .
- For $ax^4 + bx^2 + c$, substitute $u = x^2$.

Radicals and Rational Exponents

$$a^{1/n} = \sqrt[n]{a}$$

$$a^{m/n} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$$

$$\sqrt{ab} = \sqrt{a}\sqrt{b}, \quad \sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$$

Function Notation

$f(a)$ means substitute $x = a$

$$(f + g)(x) = f(x) + g(x)$$

$$(fg)(x) = f(x)g(x)$$

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} \quad g(x) \neq 0$$

Restrictions to Check

- Denominators cannot equal 0.
- Even roots require nonnegative radicands over the reals.
- Logarithm arguments must be positive.
- Squaring both sides can create extraneous solutions.